



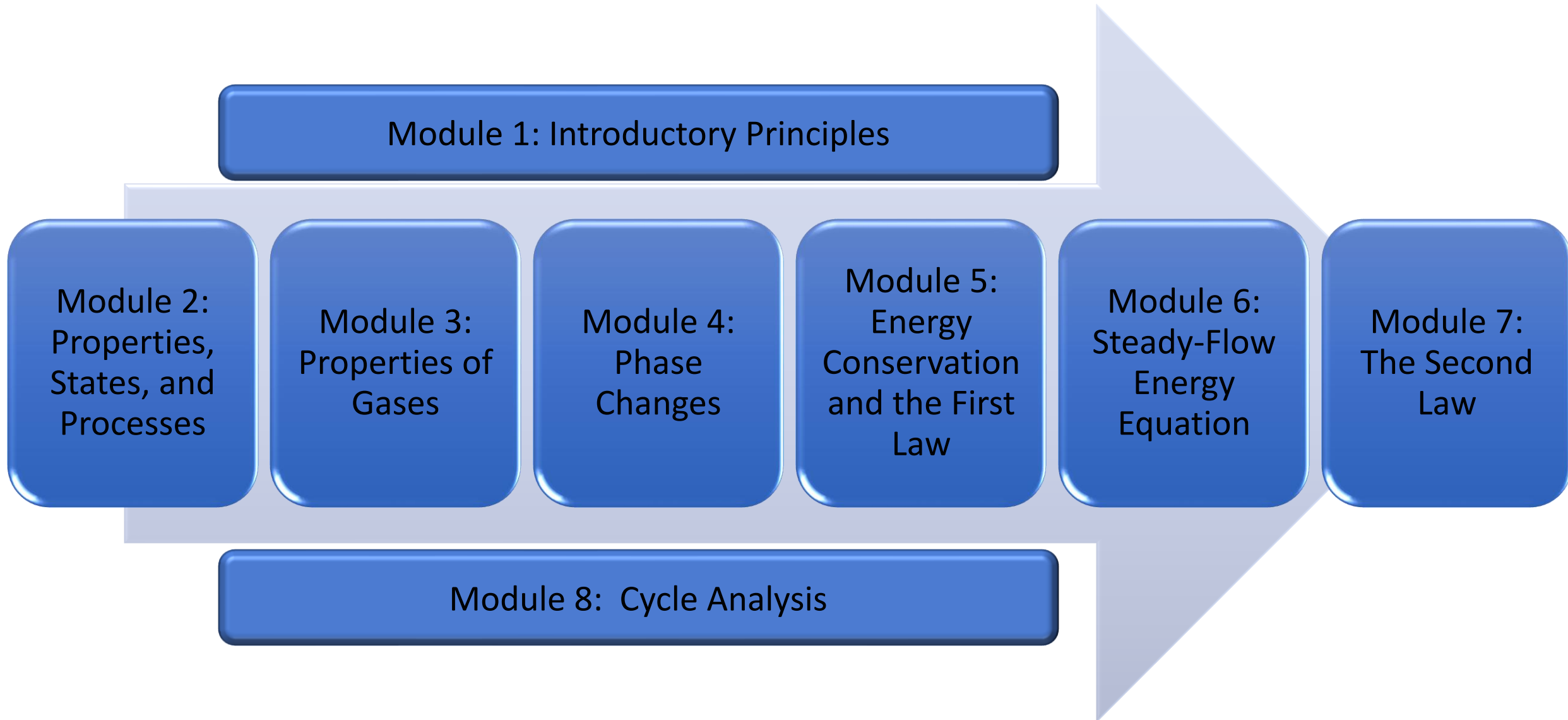
ME 300:
Engineering Thermodynamics

Overview of the Course

Goal of the course

Introduce you to concepts of energy and conservation principles, and their applications to real-world systems

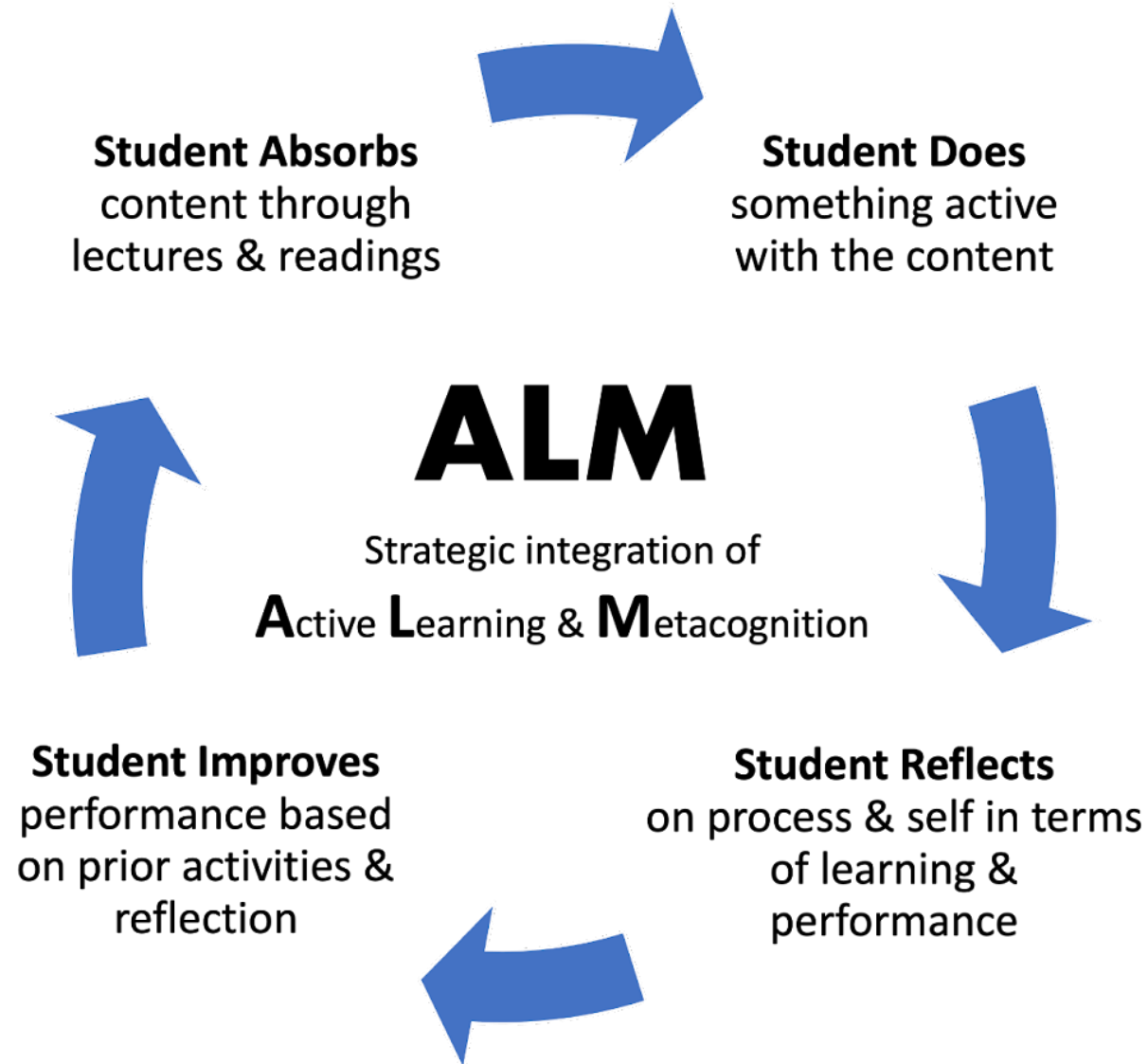
Structure of the course



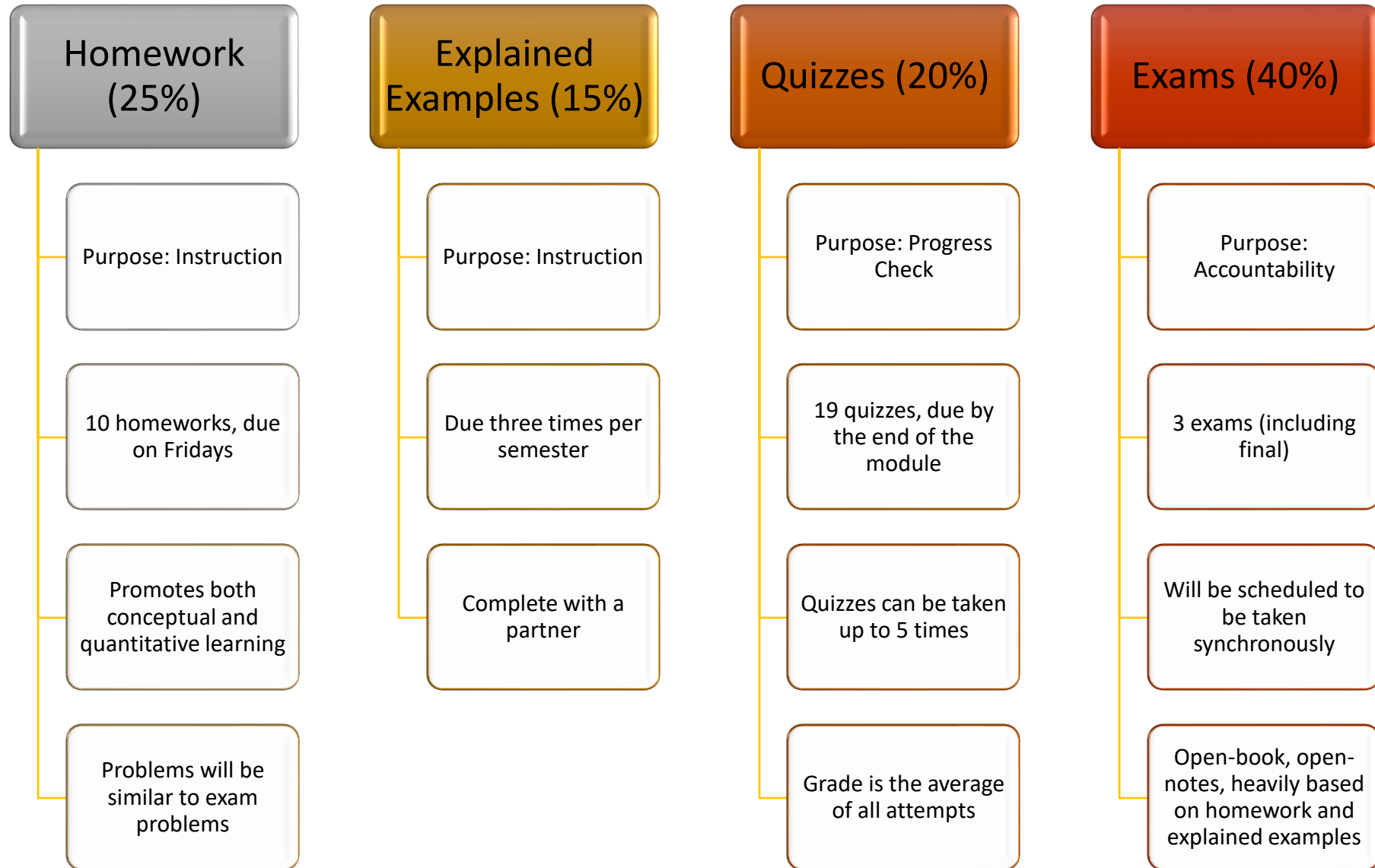
Learning Objectives

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- Obtain thermodynamic data necessary to solve thermodynamic problems and when necessary use appropriate approximations. These skills include the use of equations of state and/or tabulated property tables.
 - Write the First Law of Thermodynamics in their appropriate forms for both closed system and control volume problem.
 - Solve problems requiring First Law analysis that produce a simple single answer.
 - Make appropriate assumptions when applying the First Law to a “real-world” problem.
 - Write the Second Law of Thermodynamics in their appropriate forms for both closed system and control volume problem.
 - Apply the Second Law to determine the performance limitations of a given thermodynamic system.
 - Apply thermodynamic concepts to describe the performance of the individual components of an engineering system, e.g. a power plant, a jet engine, etc., and then relate that information to the overall performance of the entire system.

Active learning and metacognition together help students learn



Assessments



Resources

Canvas: <https://psu.instructure.com/courses/2183906>

Lecture videos

Assignments

Discussion boards

Textbook: Turns, S. R., & Pauley, L. L. (2020). Thermodynamics: concepts and applications. Cambridge University Press.

Additional videos: <https://www.cambridge.org/us/academic/subjects/engineering/thermal-fluids-engineering/thermodynamics-concepts-and-applications-2nd-edition?format=HB>

Online tools:

Ideal gas game: https://pennstate.qualtrics.com/jfe/form/SV_ehczxFmrPaJan77